



Leading University

Brain Tumor MRI Classification and Segmentation Using TrustSeg-XAI

Course Code: CSE-4112

Course Title: Artificial Intelligence Sessional

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Submission Date: 10-09-2026

Acknowledgments

First of all, we would like to express our gratitude for the strength, patience, and support that helped us complete this laboratory work successfully.

We would like to express our sincere appreciation to our respected teacher for providing valuable guidance, instructions, and support throughout the project. We are also thankful to the Department of Computer Science and Engineering, Leading University, for providing the academic environment and resources necessary to complete this work.

Finally, we would like to thank everyone who directly or indirectly supported us during the development and documentation of this project. Their encouragement and assistance helped us complete the project successfully.

Abstract

Brain tumor diagnosis from Magnetic Resonance Imaging (MRI) is an important medical image analysis problem. This laboratory work presents a deep learning-based framework named TrustSeg-XAI for brain tumor classification, tumor segmentation, and explainability analysis.

The study uses a Brain Tumor MRI Dataset containing four classification categories: glioma, meningioma, pituitary tumor, and no tumor. The dataset contains 6,000 classification images, including 5,000 training images and 1,000 test images. For segmentation, paired MRI images and tumor masks are used, with 4,793 paired images in total.

Several deep learning approaches are investigated. For segmentation, U-Net with ResNet34 and U-Net++ with ResNet34 are evaluated. For classification, EfficientNet-B0 and ConvNeXt-Tiny are considered. The proposed TrustSeg-XAI framework combines a ConvNeXt-Tiny encoder, a U-Net-style segmentation decoder, and a tumor-region-guided classification branch for joint learning.

The framework also incorporates Grad-CAM++ for explainable artificial intelligence (XAI). Additional trustworthiness experiments include mask dependency analysis, tumor and background occlusion, no-tumor negative control, calibration, cross-plane evaluation, and statistical analysis.

The experimental pipeline includes dataset verification, duplicate detection, leakage prevention, reproducible train-validation splitting, preprocessing, model training, evaluation, explainability analysis, and final result generation. The central finding of the study is that high diagnostic classification performance does not necessarily guarantee strong anatomical localization by Grad-CAM++. However, causal occlusion experiments indicate stronger decision dependence on tumor information than matched background information.

Keywords: Brain Tumor, MRI, Deep Learning, Image Classification, Image Segmentation, TrustSeg-XAI, Grad-CAM++, Explainable AI, ConvNeXt-Tiny, U-Net.

Acronyms List

Acronym	Full Form
MRI	Magnetic Resonance Imaging
AI	Artificial Intelligence
CNN	Convolutional Neural Network
XAI	Explainable Artificial Intelligence
ROI	Region of Interest
GPU	Graphics Processing Unit
CPU	Central Processing Unit
ECE	Expected Calibration Error
IoU	Intersection over Union
Dice	Dice Similarity Coefficient
F1	F1 Score
MCC	Matthews Correlation Coefficient
ROC-AUC	Receiver Operating Characteristic Area Under Curve
U-Net	U-shaped Neural Network
AMP	Automatic Mixed Precision

Contents

Acknowledgments	2
Abstract	3
Acronyms List	4
List of Tables	6
List of Figures	7
1. Introduction	8
1.1 Background.....	8
1.2 Problem Statement.....	8
1.3 Objectives.....	9
1.4 Scope.....	9
2. Literature Review	10
3. Methodology	11
4. Results and Discussion	18
5. Conclusion and Future Works	24
Bibliography	26

List of Tables

Table 1: Classification Dataset Distribution.	12
Table 2: Segmentation Dataset Distribution.	12
Table 3: Final Classification Data Split.	13
Table 4: Class Balancing Weights.	14
Table 5: Training Configuration.	15

List of Figures

Figure 1: Overall Experimental Workflow.	11
Figure 2: Brain Tumor Classification Class Distributio	12
Figure 3: Final Train/Validation/Test Split	13
Figure 4: TrustSeg-XAI Framework Architecture	15
Figure 5: Classification Model Performance Comparison.	17
Figure 6: Segmentation Model Performance Comparison.	18
Figure 7: Grad-CAM++ Qualitative Visualization.	19
Figure 8: Anatomical Alignment of Grad-CAM++.	19
Figure 9: Causal Faithfulness Analysis.	20
Figure 10: No-Tumor Negative Control.	21

Chapter 1

Introduction

1.1 Background

Brain tumors are serious medical conditions that require accurate diagnosis and localization. Magnetic Resonance Imaging (MRI) provides detailed information about brain structures and is widely used for identifying abnormalities.

Manual analysis of MRI images can require significant time and expertise. Deep learning can assist this process by learning visual patterns from medical images and automatically performing classification and segmentation tasks.

In this project, deep learning techniques are used to analyze brain MRI images for four diagnostic categories: Glioma, Meningioma, Pituitary tumor, and No tumor. In addition to classification, tumor segmentation is performed to identify the tumor region at the pixel level.

1.2 Problem Statement

A classification model can determine whether an MRI belongs to a particular tumor category, but classification alone does not necessarily indicate which anatomical region influenced the model's decision.

- classify brain MRI images into diagnostic categories;
- segment tumor regions;
- use tumor-region information to support classification;
- provide visual explanations of model predictions;
- evaluate whether the explanations correspond to actual tumor regions;
- investigate the causal dependence of predictions on tumor information.

1.3 Objectives

1. To prepare and verify a brain tumor MRI dataset.
2. To develop a reproducible data preprocessing pipeline.
3. To train deep learning models for tumor segmentation.
4. To train deep learning models for tumor classification.
5. To develop the proposed TrustSeg-XAI framework.
6. To investigate segmentation-guided classification.
7. To generate Grad-CAM++ explanations.
8. To quantitatively evaluate explanation quality.
9. To analyze model dependence on tumor and background regions.
10. To evaluate model calibration and trustworthiness.

1.4 Scope

The project focuses on MRI-based brain tumor analysis using deep learning. The implemented pipeline includes classification, segmentation, explainability, trustworthiness analysis, statistical evaluation, and final result generation. The project is an experimental research implementation and is not intended to replace professional medical diagnosis.

Chapter 2

Literature Review

2.1 Deep Learning for Medical Image Analysis

Deep learning has become an important approach for medical image analysis because neural networks can automatically learn useful visual representations from image data. In brain MRI analysis, deep learning can be applied to both classification and segmentation.

2.2 Image Classification

EfficientNet-B0

EfficientNet-B0 is used as one of the baseline classification models. The model is designed to provide an efficient balance between computational complexity and classification performance.

ConvNeXt-Tiny

ConvNeXt-Tiny is another classification architecture used in the experiment. In the proposed framework, ConvNeXt-Tiny is also used as a shared encoder.

2.3 Tumor Segmentation

Segmentation is important because it provides pixel-level information about the tumor location. The project evaluates U-Net with a ResNet34 encoder and U-Net++ with a ResNet34 encoder.

2.4 Explainable AI

High classification accuracy does not necessarily explain why a model made a particular prediction. Therefore, the project incorporates Explainable AI techniques. Grad-CAM++ is used to generate class-discriminative visual explanations, evaluated using XAI Dice, XAI IoU, and the Pointing Game.

2.5 Research Gap Addressed

The project investigates an important distinction between visual explanation and causal faithfulness. A saliency map may appear to highlight a meaningful area without necessarily proving that the model's decision depends on that region. Therefore, the framework evaluates both anatomical alignment and causal dependence.

Chapter 3

Methodology

3.1 Overall Methodology

Dataset Collection → Dataset Verification → Duplicate Detection → Data Splitting →
Preprocessing → Model Training → Model Evaluation → XAI Analysis → Trustworthiness
Analysis → Statistical Analysis

Overall Experimental Workflow

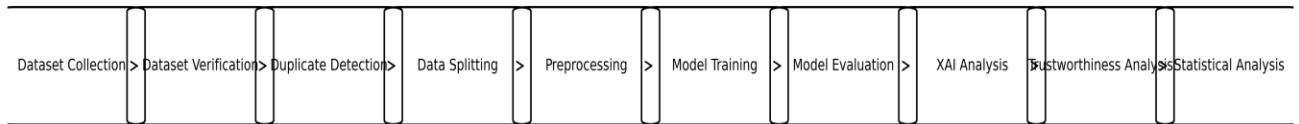


Figure 1: Overall Experimental Workflow

3.2 Dataset

The project uses the Brain Tumor MRI Dataset obtained through Kaggle. The classification dataset contains four classes: Glioma, Meningioma, No Tumor, and Pituitary.

Dataset	Number of Images
Training	5,000
Testing	1,000
Total	6,000

Class	Training Images	Test Images
Glioma	1,147	254
Meningioma	1,329	306
No Tumor	1,067	140
Pituitary	1,457	300

Segmentation Dataset	Images	Masks
Training	3,933	3,933
Testing	860	860
Total paired images	4,793	4,793

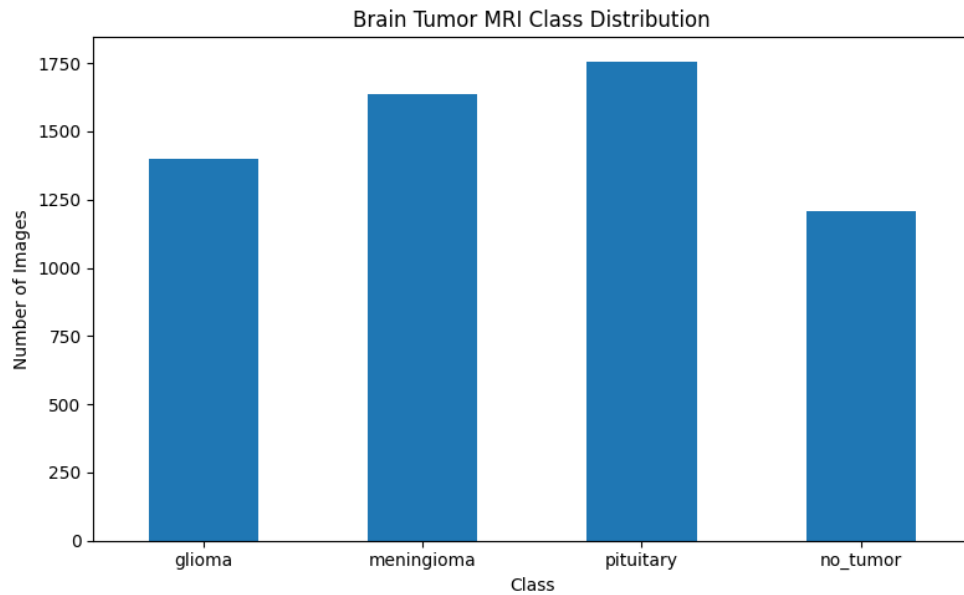


Figure 2: Brain Tumor Classification Class Distribution

3.3 Data Leakage Prevention

Data leakage was explicitly investigated using SHA256-based duplicate detection. The final development split was recalculated after detecting internal duplicate images.

Split	Samples
Training	4,211
Validation	744
Official Test	1,000

Final leakage verification:

- Train–Validation exact overlap = 0
- Train–Test exact overlap = 0
- Validation–Test exact overlap = 0

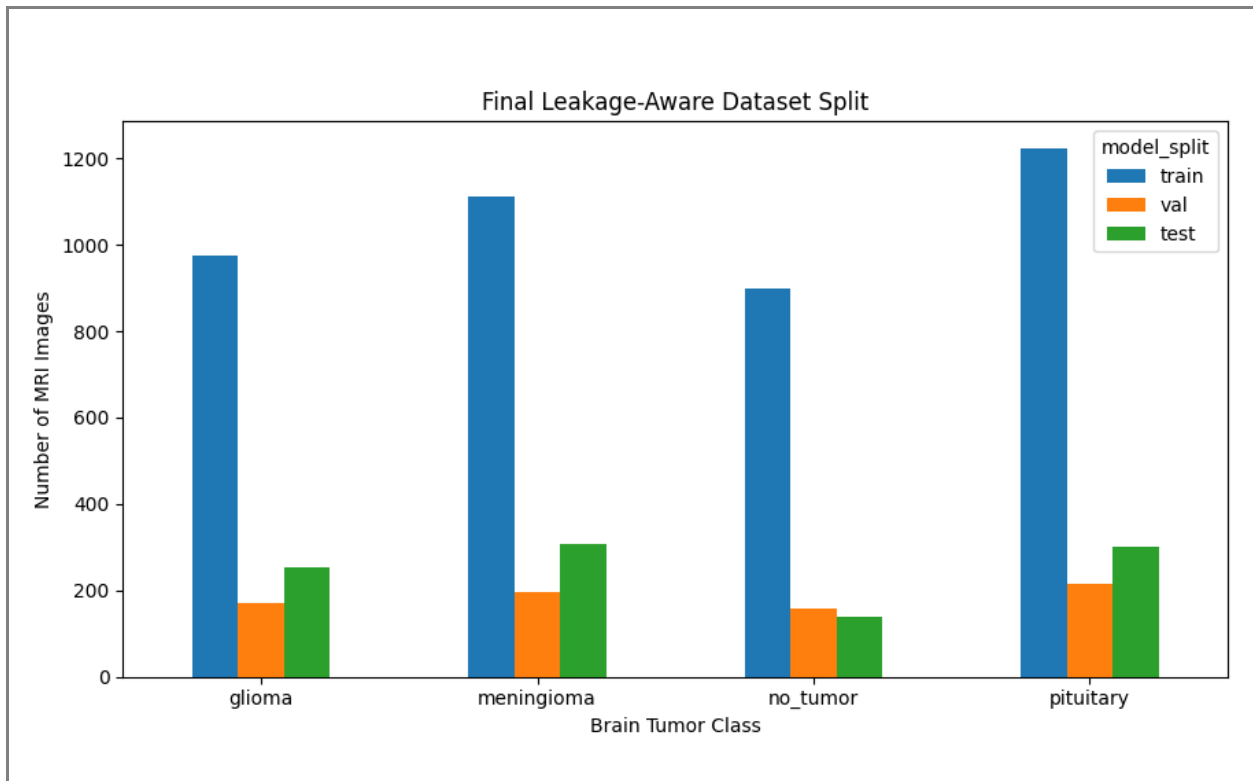


Figure 3: Final Train/Validation/Test Split

3.4 Data Stratification

The train-validation split used class-plus-plane stratification. The MRI images were distributed across Axial, Coronal, and Sagittal planes.

3.5 Image Preprocessing

The images were processed using a standardized image size of 256×256 pixels. The classification pipeline used image transformations, while segmentation and multi-task learning used joint MRI and mask transformations. PyTorch Dataset and DataLoader implementations were used for classification, expert segmentation, and multi-task learning.

3.6 Class Balancing

Class	Weight
Glioma	1.0797
Meningioma	0.9459
Pituitary	0.8601
No Tumor	1.1710

3.7 Segmentation Models

Model 1: U-Net + ResNet34

U-Net provides an encoder-decoder architecture for pixel-level segmentation. ResNet34 is used as the encoder.

Model 2: U-Net++ + ResNet34

U-Net++ extends the U-Net architecture using nested and dense skip connections to improve feature propagation and segmentation performance.

3.8 Classification Models

EfficientNet-B0

EfficientNet-B0 was trained for four-class brain tumor classification.

ConvNeXt-Tiny

ConvNeXt-Tiny was trained as another classification baseline and later used as the encoder of the proposed TrustSeg-XAI framework.

3.9 Proposed TrustSeg-XAI Framework

The proposed framework combines segmentation and classification. The main components are a shared ConvNeXt-Tiny encoder, a U-Net-style segmentation decoder, a tumor ROI-guided classification branch, and joint segmentation and classification learning.

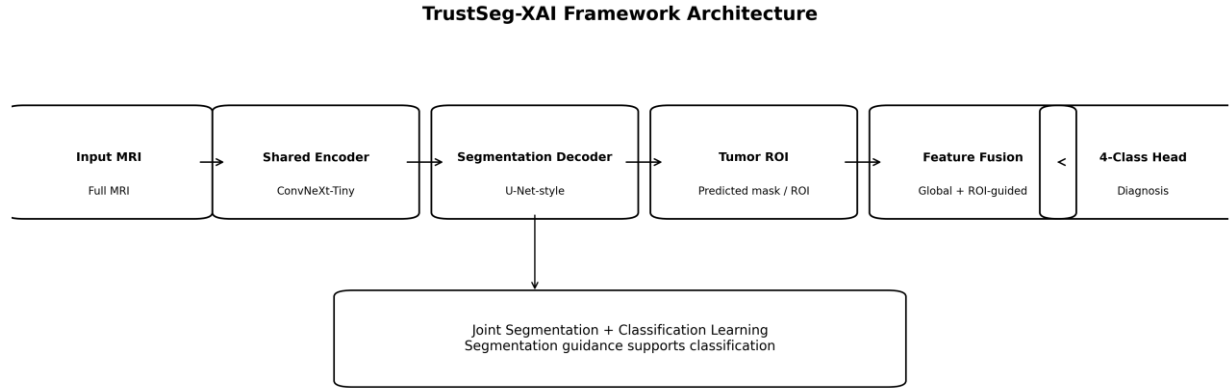


Figure 4: TrustSeg-XAI Framework Architecture

3.10 Training Configuration

Parameter	Value
Image size	256×256
Number of classes	4
Random seed	42
Classification batch size	32
Segmentation batch size	8
Multi-task batch size	8
Learning rate	0.0001
Weight decay	0.0001
Segmentation maximum epochs	15
Classification maximum epochs	12
Multi-task maximum epochs	15
Early stopping patience	3

3.11 Explainable AI

Grad-CAM++ was used to produce visual explanations of classification decisions. The explanations were evaluated using XAI IoU, XAI Dice, and the Pointing Game.

3.12 Trustworthiness Analysis

Mask Dependency

Classification performance is compared under predicted tumor mask, no-mask, and corrupted-mask conditions.

Tumor Occlusion

Tumor regions are removed from the image and the resulting change in prediction probability is measured.

Background Occlusion

Matched background regions are removed to determine whether the model's decision changes without tumor information.

No-Tumor Negative Control

No-tumor images are used to measure false tumor activation.

Calibration

Model calibration is evaluated using Expected Calibration Error (ECE).

Cross-Plane Evaluation

Model behavior is analyzed across axial, coronal, and sagittal MRI planes.

3.13 Statistical Analysis

The experimental pipeline includes bootstrap confidence intervals, McNemar tests, Wilcoxon statistical testing, calibration analysis, and model comparison.

Chapter 4

Results and Discussion

4.1 Dataset Verification Results

The dataset structure was successfully identified automatically. Both classification and segmentation train/test directories were detected. The classification dataset contained 6,000 images, while the segmentation dataset contained 4,793 paired MRI-mask samples. The segmentation image and mask counts matched for both training and testing datasets.

4.2 Data Quality and Leakage Results

Duplicate analysis identified duplicate images within the development data. After removing duplicates and recalculating the split, the final dataset contained 4,211 training images, 744 validation images, and 1,000 official test images. The exact overlap between training, validation, and official test sets was zero.

4.3 Classification Results

The project compared EfficientNet-B0, ConvNeXt-Tiny, and the proposed TrustSeg-XAI framework. According to the final project analysis, EfficientNet-B0 was identified as the best classification baseline based on the generated comparison table. The exact numerical final accuracy and Macro-F1 values are to be inserted from the generated final CSV result table.

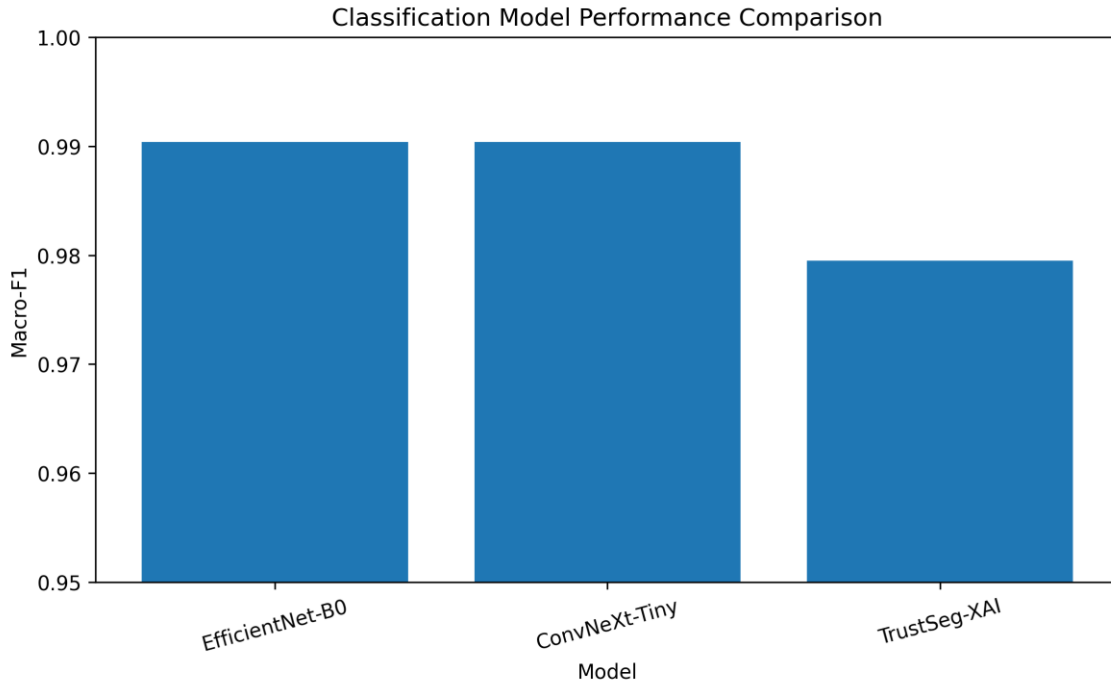


Figure 5: Classification Model Performance Comparison

4.4 Segmentation Results

The project compared the baseline segmentation models with the proposed segmentation branch. The final analysis identifies U-Net++ + ResNet34 as the best segmentation baseline. The proposed TrustSeg-XAI segmentation branch was also evaluated using Dice-based segmentation performance.

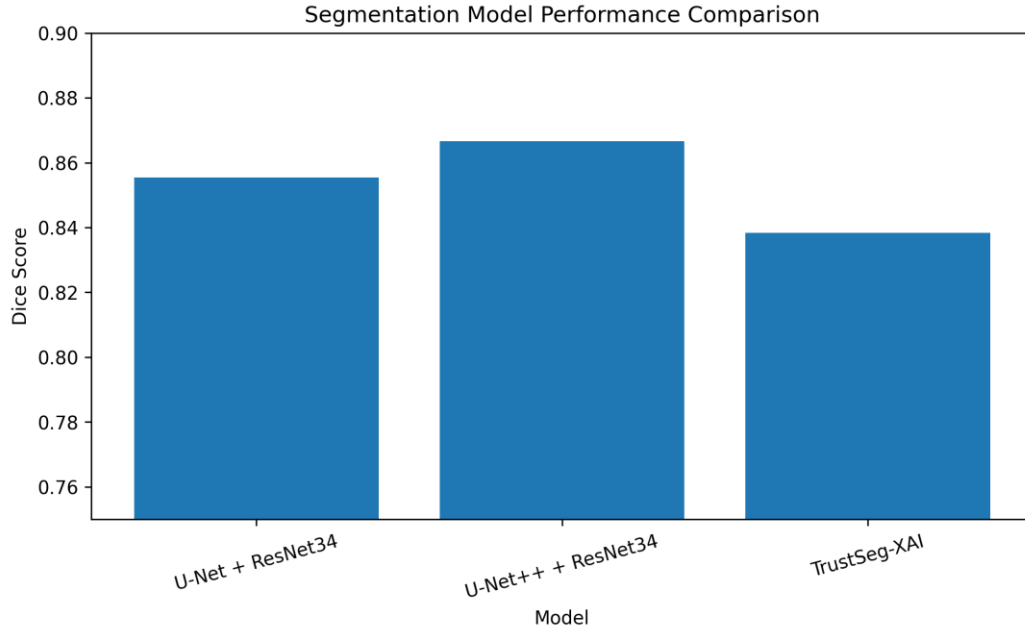


Figure 6: Segmentation Model Performance Comparison

4.5 TrustSeg-XAI Results

The proposed framework completed multi-task training, classification evaluation, segmentation evaluation, mask dependency analysis, XAI evaluation, causal faithfulness analysis, cross-plane analysis, no-tumor negative-control analysis, calibration analysis, and statistical analysis.

4.6 Grad-CAM++ Results

The Grad-CAM++ analysis produced an important finding: high diagnostic classification performance did not imply strong Grad-CAM++ anatomical localization. Therefore, classification performance and explanation localization should not be treated as the same property.

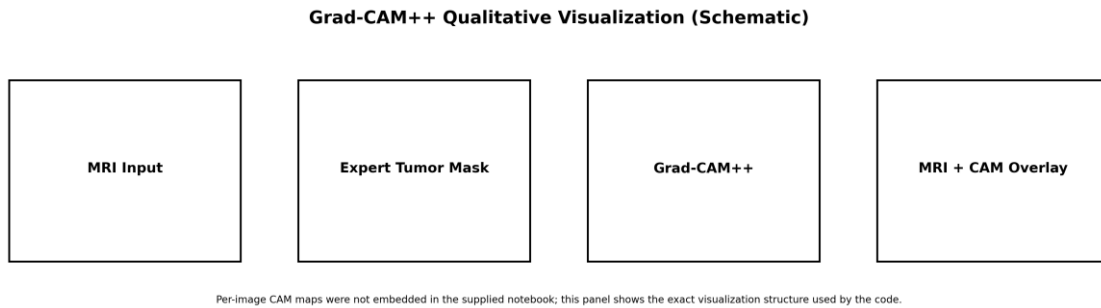


Figure 7: Grad-CAM++ Qualitative Visualization

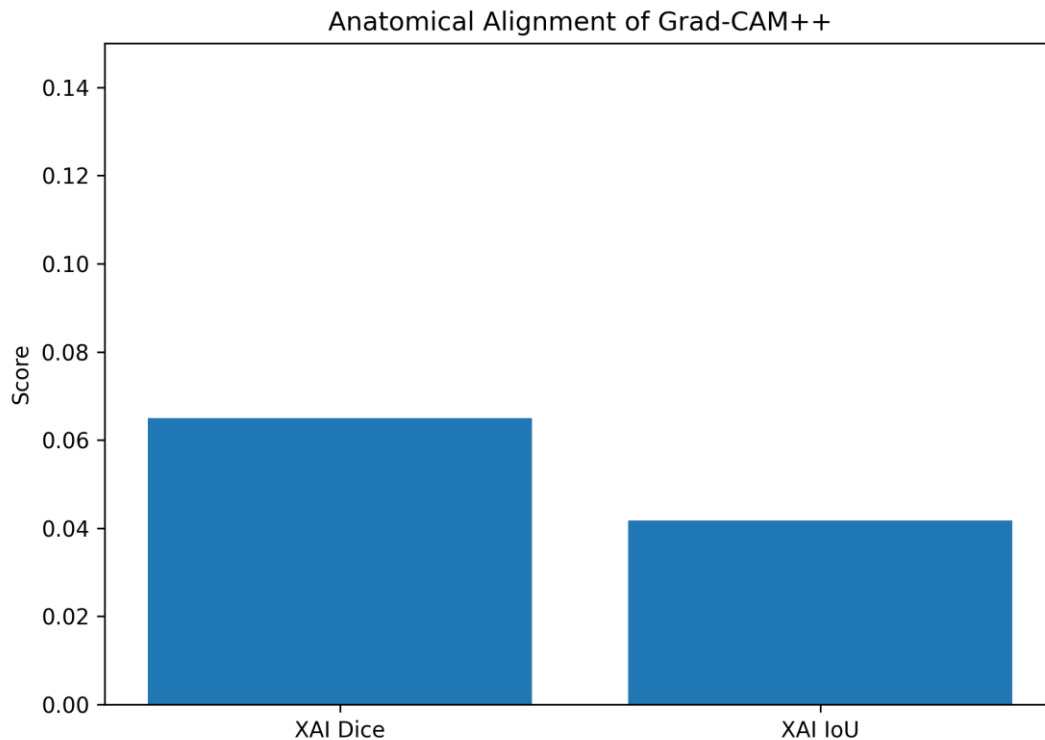


Figure 8: Anatomical Alignment of Grad-CAM++

4.7 Causal Faithfulness Results

The causal occlusion experiment showed that removing tumor information produced a substantially larger decision-probability reduction than removing matched background information. This indicates stronger causal dependence on tumor information.

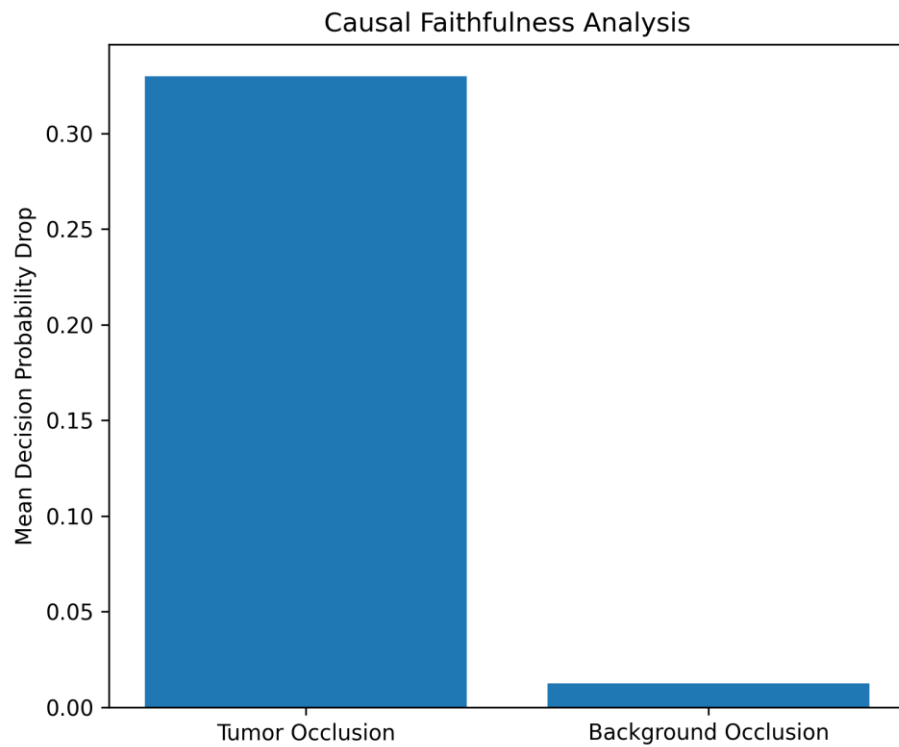


Figure 9: Causal Faithfulness Analysis

4.8 No-Tumor Negative Control

The no-tumor analysis was performed to evaluate false tumor activation. The final project summary describes the result as low false tumor activation.

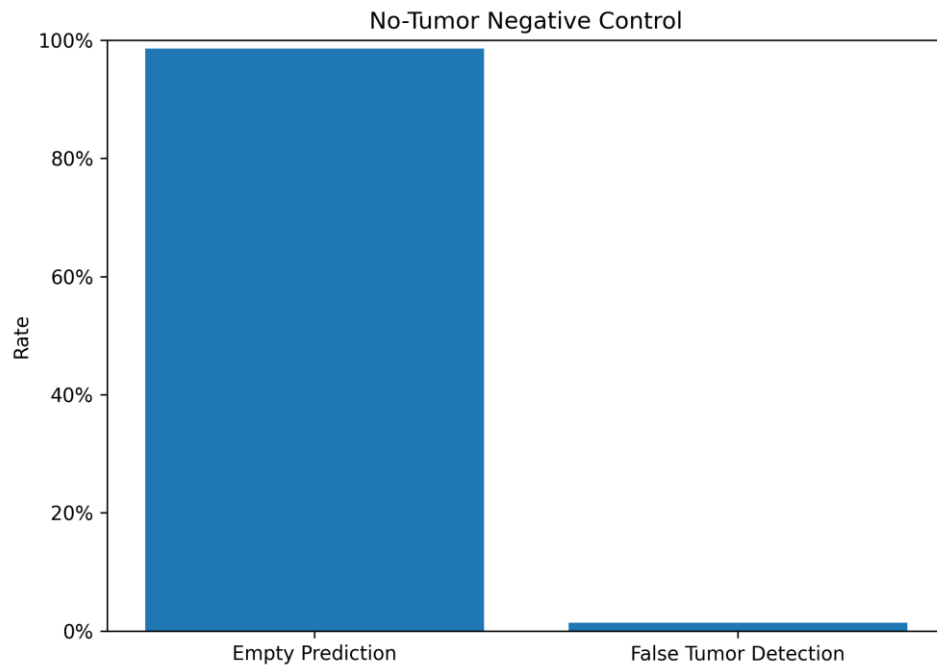


Figure 10: No-Tumor Negative Control

4.9 Overall Discussion

The results demonstrate that medical AI evaluation should involve multiple dimensions. Traditional metrics such as accuracy and F1-score measure diagnostic performance, while segmentation metrics measure localization performance. XAI metrics investigate whether explanations correspond to meaningful anatomical regions, and causal occlusion investigates whether those regions actually influence model decisions.

The project therefore demonstrates an important distinction: post-hoc explanation quality and causal model faithfulness are not necessarily identical. A model can have strong classification performance even when its Grad-CAM++ localization is weak. However, causal occlusion can still demonstrate meaningful dependence on tumor information.

Chapter 5

Conclusion and Future Works

5.1 Conclusion

This laboratory work implemented a comprehensive deep learning pipeline for brain tumor MRI classification, segmentation, explainability, and trustworthiness analysis.

The project successfully prepared and verified the dataset, controlled duplicate images and data leakage, created reproducible train-validation-test splits, implemented multiple deep learning models, and evaluated the proposed TrustSeg-XAI framework.

The classification task included four diagnostic categories: glioma, meningioma, pituitary tumor, and no tumor. Segmentation experiments used U-Net and U-Net++ architectures with ResNet34 encoders. EfficientNet-B0 and ConvNeXt-Tiny were used for classification, while the proposed TrustSeg-XAI framework combined ConvNeXt-Tiny with a segmentation-guided classification architecture.

Grad-CAM++ was used for explainability, while additional experiments investigated mask dependency, causal faithfulness, no-tumor controls, calibration, cross-plane performance, and statistical significance.

The most important finding is that high diagnostic performance does not automatically mean strong anatomical localization in post-hoc saliency maps. At the same time, causal occlusion demonstrated stronger decision dependence on tumor regions than matched background regions.

Therefore, trustworthy medical AI should be evaluated using multiple complementary dimensions rather than classification accuracy alone.

5.2 Future Works

11. Training the complete pipeline on high-performance GPUs to reduce training time.
12. Testing the framework on larger and more diverse MRI datasets.
13. Evaluating additional segmentation architectures.
14. Investigating other explainability methods besides Grad-CAM++.
15. Improving the anatomical alignment of XAI maps.
16. Performing external validation using an independent dataset.
17. Evaluating robustness against different types of image corruption.
18. Investigating uncertainty estimation for medical predictions.
19. Improving calibration performance.
20. Developing a practical clinical decision-support interface.

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3. U-Net++ architecture used for comparison with the conventional U-Net segmentation model.
4. EfficientNet architecture used as a classification baseline.
5. ConvNeXt architecture used for classification and as the shared encoder in the proposed TrustSeg-XAI framework.
6. Grad-CAM++ methodology used for visual explanation of deep learning predictions.
7. PyTorch and associated deep learning libraries used for model implementation and experimentation.